

Plagiarism Detection in Handwritten Assignments using OCR and Cosine Similarity Analysis

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Abstract:

The research paper addresses the pressing need for a robust plagiarism detection system tailored specifically for handwritten assignments submitted in educational environments. Leveraging Optical Character Recognition (OCR) technology, handwritten text is converted into digital format, enabling the application of conventional plagiarism detection techniques. The cosine similarity analysis method is employed to calculate similarity scores between submitted assignments, facilitating the identification of potentially plagiarized content.

Furthermore, recognizing the importance of accessibility and usability, this research integrates Flask and React frameworks for the development of a user-friendly web application. By employing these technologies, the plagiarism detection system becomes readily accessible to both educators and students, streamlining the process of submission, analysis, and feedback.

This paper aims to explore the technical intricacies of implementing such a system, evaluating its effectiveness in detecting plagiarism within handwritten assignments. Additionally, it examines the implications of integrating OCR and cosine similarity analysis into the educational landscape, discussing the potential benefits and challenges associated with adopting this technology-driven approach to uphold academic integrity. The research work contributes to the ongoing discourse on plagiarism detection methods, fostering a culture of academic honesty and accountability in

educational institutions.

Keywords: Cosine Similarity Analysis, Optical Character Recognition(OCR).

I. INTRODUCTION:

Plagiarism, as defined by the Oxford Dictionary, refers to the act of appropriating someone else's work, ideas, or thoughts and presenting them as one's own. The term originates from the Latin word "Plagiarus," which conveys the notions of kidnapping or plundering. "Plagium," derived from "Plaga," meaning to capture or ensnare, denotes a kidnapper. In contemporary usage, plagiarism pertains to the act of plagiarizing. The process of scrutinizing works or documents for plagiarism is termed Plagiarism Detection or Checker.

The history of plagiarism traces back to religious texts, many of which were devoid of known authors, leading to extensive copying and integration into subsequent works. By the mid-1600s, accusations of plagiarism were rampant across various creative fields. In 1709, the first copyright law was enacted, initially aimed at safeguarding publishers' rights rather than authors'. Subsequent legislation, however, focused on protecting authors' rights. James Boswell, renowned for his biography of Samuel Johnson, was a lawyer who contested the duration of authors' copyrights, which at the time extended to 21 years.

Academic integrity stands as a cornerstone of scholarly pursuit, fostering an environment where originality, creativity, and rigor thrive[1]. Yet, in the digital age, the proliferation of information and ease of access to resources have also amplified the challenge of maintaining this integrity. Plagiarism, the unauthorized use or representation of another's work as one's own, poses a significant threat to the integrity of academic institutions worldwide. In response to this imperative, this research endeavors to present a multi-technique approach to academic plagiarism detection, aiming to enhance the efficacy and accuracy of current detection methodologies[2]. Central to this approach is the integration of various techniques, each offering unique insights into different facets of plagiarism. Digital pattern matching provides a structural analysis of submitted documents, identifying similarities that may indicate direct copying. Optical Character Recognition (OCR) technology enables the extraction of text content from documents, laying the groundwork for textual analysis. Cosine similarity analysis further extends this analysis by quantifying the textual resemblance between submissions. Moreover, the integration of Flask and

React frameworks enables the development of a user-friendly web application, enhancing accessibility and usability for educators and students alike.

Through experimental validation on diverse datasets, this research seeks to demonstrate the efficacy and robustness of the proposed methodology in detecting various forms of academic misconduct.

II. TYPES OF PLAGIARISM

A. Plagiarism of duplicate assignments

Plagiarism of duplicate assignments is a pervasive issue in educational settings, posing significant ethical concerns and undermining the integrity of academic assessment. This phenomenon involves the submission of identical or highly similar assignments by different individuals, without proper attribution or acknowledgment of the original source. Such actions not only violate academic integrity policies but also diminish the educational value of assignments by thwarting the assessment of students' genuine comprehension and analytical abilities.

B. Plagiarism of same handwriting assignments

Plagiarism of assignments with identical handwriting involves submitting work copied from others without attribution. It subverts academic integrity, negates originality, and deceives educators. This misconduct undermines the assessment process and devalues genuine learning efforts. Preventing such plagiarism requires educators to be vigilant in detecting handwriting inconsistencies, employing plagiarism detection tools, and educating students on ethical conduct. Students must recognize the importance of producing their own work, refrain from copying, and seek assistance when needed.

C. Plagiarism of similar text assignments

Plagiarism of similar text assignments occurs when individuals submit work that closely resembles existing content without proper citation or acknowledgment. This form of academic dishonesty undermines the principles of originality, integrity, and fairness in education. By copying or paraphrasing without attribution, students bypass the critical thinking and analytical skills that assignments aim to assess. Moreover, it creates an unfair advantage for those who engage in plagiarism, as they receive credit without demonstrating genuine understanding or effort.

III. COSINE SIMILARITY ANALYSIS

Cosine similarity, ranging between 0 and 1, measures the similarity between two non-zero vectors, A and B, by calculating the cosine of the angle between them as shown in Fig 1. When the angle between the vectors is 90 degrees, the cosine similarity is 0, indicating orthogonality or perpendicularity. As the cosine similarity approaches 1, it signifies a smaller angle between vectors A and B. This relationship is visually represented in the accompanying images [3]. Two vectors exhibit a cosine angle similarity of 98 percent.

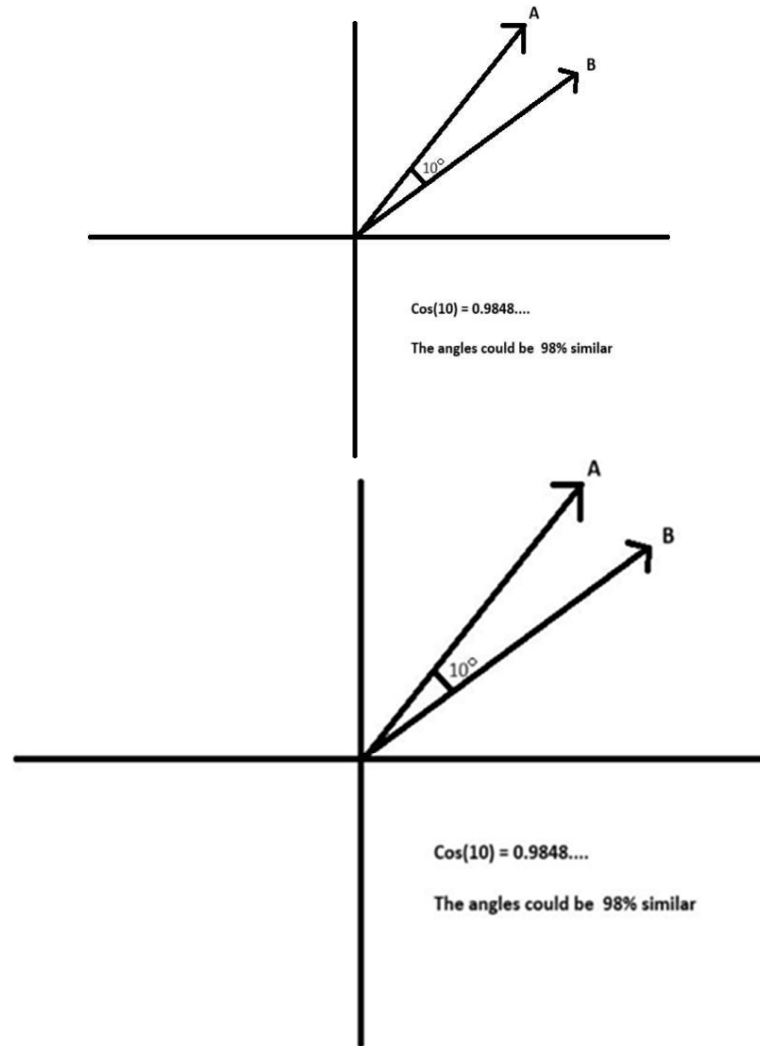


Fig 1. Plot of Cosine Similarity

In the proposed plagiarism detection system, cosine similarity analysis serves as a pivotal component for evaluating the likeness between handwritten assignments. Following the conversion of handwritten content into digital text through OCR, cosine similarity analysis quantifies the degree of resemblance between assignments by computing the cosine of the angle between their vector representations in a high-dimensional space. This method enables

the generation of similarity scores, which offer a nuanced understanding of the extent of similarity between documents. By establishing a threshold cosine similarity score, the system can effectively discern between acceptable similarities, such as shared vocabulary or common phrases, and instances of plagiarism [4].

IV. OPTICAL CHARACTER RECOGNITION

Optical Character Recognition (OCR) technology serves as the cornerstone of the proposed plagiarism detection system for handwritten assignments, offering a transformative solution to the inherent challenges posed by handwritten content. Fig 2. shows the Flowchart of OCR. By seamlessly converting analog text into digital format, OCR enables the integration of handwritten submissions into the digital realm, where traditional plagiarism detection algorithms can be applied [5]. OCR streamlines workflow processes, enabling educators to efficiently analyze and compare numerous submissions for potential instances of plagiarism.

Furthermore, the adaptability of OCR systems ensures their effectiveness across various handwriting styles and languages, thereby catering to the diverse needs of students and educators. This adaptability is particularly valuable in educational environments characterized by linguistic and cultural diversity, where a standardized approach to plagiarism detection may fall short.

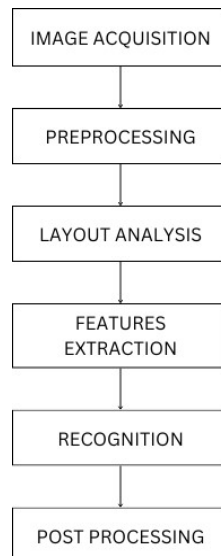


Fig 2. Flowchart of OCR

V. METHODOLOGY

A. *Collection of data (IAM dataset) for different writings*

The IAM Handwriting Database is a crucial resource for handwriting recognition. With over 1,100 pages of handwritten text from 657 writers, it offers a comprehensive dataset that covers a wide spectrum of writing styles, languages, and content. Notably, IAM provides ground truth transcriptions for each handwritten sample, facilitating meticulous training and evaluation of handwriting recognition systems. Its strength lies in its ability to mirror real-world variability, including variations in writing styles, line spacing, and text content, making it an ideal resource for those seeking to advance their understanding of handwriting recognition.

B. *Training of the data using CNN for conversion*

Neural networks have emerged as powerful tools in the realm of handwriting recognition. Convolutional Neural Networks (CNNs) excel in extracting hierarchical features from handwritten images, enabling them to recognize individual characters or entire words. Additionally, Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are highly effective in handling sequential data like handwriting. These networks capture temporal dependencies and transcribe text from sequences of data points representing pen strokes. The hybrid models that combine CNNs and RNNs provide robust recognition systems, and attention-based mechanisms, like the Transformer architecture, enhance adaptability during recognition.

C. *Conversion of image to text using OCR*

Optical Character Recognition (OCR) plays a pivotal role in converting handwritten or printed text into machine-readable form. The OCR process involves image acquisition, preprocessing, segmentation, feature extraction, pattern recognition, and post-processing. In the context of the IAM Handwriting Database, OCR systems can greatly benefit from IAM's challenging real-world variations. This resource allows OCR systems to improve their capabilities in handling diverse writing styles, languages, and content. By integrating OCR with neural networks, such as CNNs and RNNs, handwriting recognition accuracy can be significantly enhanced. This synergy has far-reaching implications, enabling applications in text digitization, document indexing, and automated transcription services, setting new standards in the field [6].

D. *Plagiarism detection using Cosine similarity analysis*

Plagiarism detection using cosine similarity analysis is a common technique employed in Natural language processing (NLP) and text mining [7]. Cosine similarity measures the

cosine of the angle between two vectors, which represent the frequency of occurrence of words in the documents being compared. Here's how it works:

1. **Text Preprocessing:** First, the text documents need to be preprocessed. This involves steps such as tokenization (splitting the text into words or phrases), removing stopwords (commonly occurring words like "the", "is", "and", etc.), and stemming or lemmatization (reducing words to their base or root form).
2. **Vectorization:** After preprocessing, the text documents are converted into numerical vectors. One common approach is the term frequency-inverse document frequency (TF-IDF) vectorization. This method assigns weights to each term in the document based on its frequency within the document and across all documents in the corpus.
3. **Cosine Similarity Calculation:** Once the documents are represented as vectors, cosine similarity is calculated between pairs of documents using their vector representations. Cosine similarity ranges from -1 to 1, where 1 means the documents are identical, 0 means they are completely dissimilar, and -1 means they are exactly opposite.
4. **Thresholding:** Based on the cosine similarity scores obtained, a threshold is set to determine whether the documents are plagiarized or not. If the cosine similarity score exceeds the threshold, it indicates a high degree of similarity between the documents and raises suspicion of plagiarism.
5. **Evaluation and Analysis:** Finally, the results of the plagiarism detection algorithm are evaluated, and further analysis may be conducted to verify the detected instances of plagiarism.

VI. RESULTS

The different handwritten texts are stored in file1, file2, file3 and file4 as shown below in Fig 3,4,5,6.

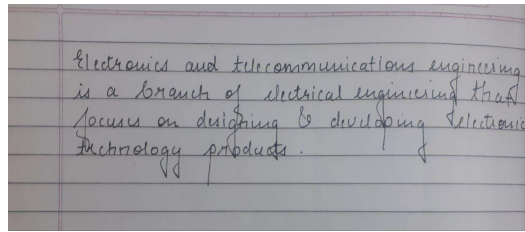


Fig 3. Contents of file1

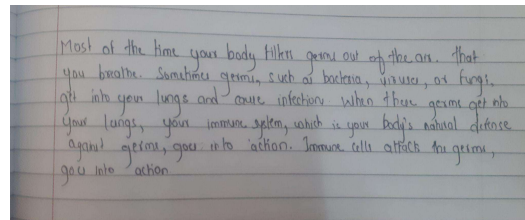


Fig 4. Contents of file2

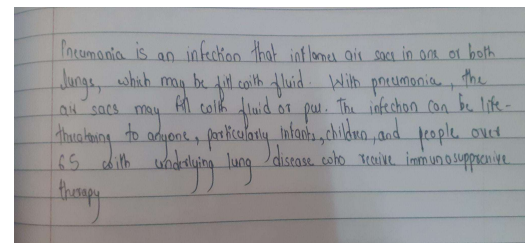


Fig 5. Contents of file3

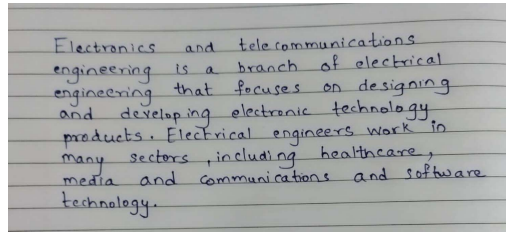


Fig 6. Contents of file4

Using the Cosine similarity analysis, each file is compared with others and the similarity index is calculated. file1 is compared with file2, file3 and file4 as shown in Table 1. It is evident that the similarity index is more for file1 & file4 compared to file1 & file2 and file1 & file3. Similarly, file2 is compared with file1, file3 and file4 as shown in Table 2. Comparison results of file3 with file1, file2 and file4 as shown in Table 3. Comparison results of file4 with file1, file2 and file3 as shown in Table 4.

Table 1: Comparison of file1 with remaining files

Sr.No.	File Names	Similarity
1	file1 vs file2	5.62%
2	file1 vs file3	2.59%
3	file1 vs file4	53.9%

Table 2: Comparison of file2 with remaining files

Sr.No.	File Names	Similarity
1	file2 vs file1	5.62%
2	file2 vs file3	8.47%
3	file2 vs file4	5.35%

Table 3: Comparison of file3 with remaining files

Sr.No.	File Names	Similarity
1	file3 vs file1	2.59%
2	file3 vs file2	8.47%
3	file3 vs file4	4.97%

Table 4: Comparison of file4 with remaining files

Sr.No.	File Names	Similarity
1	file4 vs file1	53.9%
2	file4 vs file2	5.35%
3	file4 vs file3	4.97%

VII. CONCLUSION

The project titled "Plagiarism Detection in Handwritten Assignments using OCR and Cosine Similarity Analysis" embodies a transformative exploration into safeguarding intellectual integrity and creativity. In a time marked by abundant access to information and effortless replication, the demand for effective plagiarism detection mechanisms is paramount.

The project has illuminated the compelling reasons driving its inception, highlighting the importance of preserving intellectual property and thwarting unethical content duplication. It has outlined the methodologies underpinning these systems, ranging from sophisticated text analysis to feature extraction, enabling accurate pinpointing of plagiarism instances.

This project holds significance far beyond academia; it embodies the fundamental principles of authenticity and originality in journalism, research, and creative content generation. For educators, it simplifies the task of upholding academic integrity, ensuring that the educational journey remains a stronghold of trust and ethics. In journalism, it bolsters the credibility of publications by preventing unintentional or intentional content duplication. In research, it preserves the integrity of scientific discoveries and the dissemination of knowledge.

Looking ahead, the project serves as a guiding light in the ever-evolving landscape of plagiarism detection. As technology progresses, it presents promising opportunities for integrating even more advanced tools, machine learning models, and extensive databases. With user-friendly interfaces and integration across various platforms, this technology can be leveraged to cultivate a culture of ethical content creation, where originality and intellectual integrity are not merely ideals but living principles vital for the advancement and ethics of our interconnected society. The road ahead is promising, and the possibilities are boundless, making this project an essential stride toward a more authentic and trustworthy digital world.

VIII. FUTURE SCOPE

In addition to the aforementioned technical advancements, future research should also prioritize addressing broader educational and ethical considerations. Collaboration with educators, students, and educational policymakers is essential to ensure that the development and implementation of the plagiarism detection system align with pedagogical goals and ethical standards. Furthermore, exploring the socio-cultural implications of plagiarism detection technology, particularly in diverse educational contexts, can inform the development of more inclusive and culturally sensitive approaches [8].

Moreover, efforts to raise awareness about academic integrity and plagiarism prevention should

be integrated into educational curricula at all levels. Providing students with clear guidelines on proper citation practices and fostering a culture of originality and critical thinking can mitigate the prevalence of plagiarism.

Ultimately, the future scope of this research extends beyond technical innovation to encompass a holistic approach that considers the complex interplay between technology, education, and ethics. By remaining attuned to the evolving needs of educational stakeholders, we can continue to advance towards a more integrity-driven academic environment.

REFERENCES

1. MAC Jiffriya MAC Akmal Jahan, Roshan G. Ragel, "Plagiarism Detection on Electronic Text based Assignments using Vector Space Model ," IEEE, 2014.
2. Nikhil Paymode, Rahul Yadav, Sudarshan Vichare, Suvarna Bhoir, "Online Assignment Plagiarism Detector, " International Journal of Advanced Research in Science, Communication and Technology, vol.4, issue.2, April 2021.
3. Mohamed A. El-Rashidy, Ramy G. Mohamed, Nawal A. El-Fishawy, Marwa A. Shouman, "An effective text plagiarism detection system based on feature selection and SVM techniques," Multimedia Tools and Applications(Springer), 2024.
4. Hiten Chavan, Mohd. Taufik, Rutuja Kadave, Nikita Chandra, "Plagiarism Detection Using Machine Learning ," International Journal of Research in Engineering, Science and Management, vol.4, issue.4, April 2021.
5. El Mostafa Hambi, Faouzia Benabbou, "A New Online Plagiarism Detection System based on Deep Learning, " International Journal of Advanced Computer Science and Applications, vol.11, no.9, 2020.
6. Mohamed A. El-Rashidy, Ramy G. Mohamed, Nawal A. El-Fishawy, Marwa A. Shouman, "Reliable plagiarism detection system based on deep learning approaches," Neural Computing and Applications, 2022.
7. Aparna Babu, Anju S, Archana P Udayan, Giny Mary John, Divya S B, "Online Plagiarism Detection and Result Evaluation using Data Mining and NLP," International Journal of Engineering Research and Technology, vol.10, issue.4, 2022.
8. Niraj Mohabey, Yash Gavanang, Abubakkar Khan, Lavesb Singh Chib, Bhushan Patil, "Plagiarism Detection for project report using machine learning," International Journal of Engineering Technology and Management Sciences, issue.3, vol.7, May-June2023.